IC-272 — Introduction to Machine Learning

Lab 8 — Naive Bayes

Indian Institute of Technology Mandi

Total marks: 100 — Duration: 2 hours

Marks: 20 for code, 80 for the viva. There are three tasks. Your TA takes a short viva on each task *as soon as you finish it* — do not save them all for the end.

There is also a homework section at the end. It will not be vivaed and carries no marks today. It is examinable in the final lab exam.

You will reuse `design_matrix` (Lab 4), `standardise` (Lab 5), `sigmoid` and `fit_logistic` (Lab 7), your confusion-matrix and metric functions (Lab 1), and the two ROC functions from the Lab 7 homework, `roc_curve_scratch` and `auc_trapezoid`. Copy them into a cell at the top. **If you did not finish the Lab 7 homework, you must write those two functions now** — Task 3 cannot be done without them, and the time comes out of this lab.

Rules

- Permitted: `numpy`, `pandas`, `matplotlib`. **`scikit-learn` and `scipy.stats` are not permitted** — a task that calls either scores zero for that task. Write the Gaussian density yourself.

Setting up — type this first

`ic272_lab8_loans.csv` is the Lab 7 loan data — the same 500 applicants in the same order — with one extra column, `employment`, which you will need only for the homework. Today's tasks use `income_k` and `utilisation`, exactly as in Lab 7.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("ic272_lab8_loans.csv")
X   = df[["income_k", "utilisation"]].to_numpy(float)
emp = df["employment"].to_numpy()
y   = df["defaulted"].to_numpy(int)

idx = np.random.default_rng(0).permutation(len(y))
te, tr = idx[:100], idx[100:]
X_train, y_train, emp_train = X[tr], y[tr], emp[tr]
X_test,  y_test,  emp_test  = X[te], y[te], emp[te]
```

The split is identical to Lab 7, so the logistic regression you fit last week is the model you will compare against today.

Why this lab exists — read this, then run it

Logistic regression models $P(y \mid x)$ directly. Naive Bayes goes the other way round: it models how each class *generates* its features, $P(x \mid y)$, and then turns that around with Bayes' rule,

$$P(y \mid x) = \frac{P(x \mid y)\,P(y)}{P(x)}.$$

$P(x)$ is the same for every class, so to compare classes you only need $P(x \mid y)\,P(y)$.

The "naive" part is the assumption that, *within a class*, the features are independent of one another, so

$$P(x \mid y) = \prod_{\alpha=1}^{d} P(x_\alpha \mid y).$$

That turns one hard d-dimensional estimate into d easy one-dimensional ones. For continuous features the lecture models each one as a Gaussian with its own mean $\mu_{\alpha c}$ and variance $\sigma^2_{\alpha c}$ per class:

$$P(x_\alpha \mid y = c) = \frac{1}{\sqrt{2\pi\sigma^2_{\alpha c}}} \exp\left(-\frac{(x_\alpha - \mu_{\alpha c})^2}{2\sigma^2_{\alpha c}}\right).$$

Why everything happens in logs

A product of many probabilities is a very small number. Run this:

```
p = np.full(400, 0.01)          # 400 probabilities, each 0.01
print(np.prod(p))               # the product
print(np.sum(np.log(p)))        # the sum of the logs
```

The product prints 0.0. The true value, 10^{-800}, is far smaller than the smallest number a computer can store, so it underflows to zero and every class would look equally impossible. The sum of the logs, -1842.07, is perfectly ordinary. So, as the lecture shows, we compare classes using

$$\log P(y = c) + \sum_{\alpha=1}^{d} \log P(x_\alpha \mid y = c),$$

which is the same idea as the log-likelihood in Lab 7.

Task 1 — Fitting Gaussian Naive Bayes [code 6 — viva 24]

(a) Write `gaussian_pdf(x, mu, var)`, vectorised. **[2]**

(b) Write `fit_gaussian_nb(X, y)` returning three arrays: the class priors (shape $(2,)$), the per-class feature means (shape $(2, d)$), and the per-class feature variances (shape $(2, d)$). Use the lecture's estimates — divide the variance by n_c, not $n_c - 1$. **[3]**

(c) Fit it on the training set and print all three arrays. **[1]**

Check as you go using the worked example from the lecture: class A heights $[160, 170, 165]$, class B heights $[150, 155, 145]$, equal priors, test height 162. Your functions should reproduce $P(162 \mid A) \approx 0.0746$, $P(162 \mid B) \approx 0.0013$ and a posterior of about 0.98 for class A. (The slide shows 0.0747 because it rounds its intermediate steps.) Get this right before calling your TA.

Task 2 — Prediction in log space [code 8 — viva 32]

(a) Write `predict_proba_nb(X, priors, means, variances)` returning $P(y = 1 \mid x)$ for every row. Compute the log-score for each class as above — **do not multiply densities together** — then convert the two log-scores into a probability. **[5]**

(b) Predict on the test set, threshold at 0.5, and report accuracy, precision, recall and F_1 with your Lab 1 functions. **[3]**

For the conversion: if L_0 and L_1 are the two log-scores, then $P(y = 1 \mid x) = \dfrac{e^{L_1}}{e^{L_0} + e^{L_1}}$. Divide top and bottom by e^{L_1} and you get a function you wrote last week — use it, rather than exponentiating the raw scores, which would bring the underflow straight back.

Task 3 — Naive Bayes against logistic regression [code 6 — viva 24]

(a) Fit your Lab 7 logistic regression on the same training rows (standardised, as in Lab 7) and get its test probabilities. **[1]**

(b) Using `roc_curve_scratch` and `auc_trapezoid` from the Lab 7 homework, compute the AUC of both models on the test set. **[3]**

(c) Plot both ROC curves on one axes, with the diagonal for reference. **[2]**

One model learns $P(y \mid x)$ directly; the other learns how each class generates its data and works backwards. Before calling your TA, decide which you expected to win and why, then compare with what you actually got.

Homework — not vivaed today, examinable in the final lab exam

Complete this homework before the next lab. The functions you write here are reused later in the course and will be called by the names given below. If you skip them now, you will be writing them under time pressure later.

H1. Categorical Naive Bayes and Laplace smoothing. Write `fit_categorical_nb(e, y, l)` returning, for each class, the probability of each employment category, using the lecture's estimate

$$\hat{\theta}_{jc} = \frac{\#\{\text{class } c \text{ with category } j\} + l}{\#\{\text{class } c\} + l\,K},$$

with $K = 4$ categories. Fit it on the training rows with $l = 0$ and with $l = 1$, and print both tables. One entry of the $l = 0$ table is exactly zero. Which, and why?

H2. The zero-probability problem, measured. Combine the two models: for each class, add the categorical log-probability of the applicant's employment to the Gaussian log-score from Task 2. Find the test applicants who are students *and* defaulted, and print their $P(\text{default})$ from the numeric-only model, the combined model with $l = 0$, and the combined model with $l = 1$. Compare the three AUCs as well. Explain what a single zero count in the training data did to those applicants, and why $l = 1$ fixes it.

H3. Breaking the naive assumption. Make a feature matrix with `utilisation` copied in three times, so the model sees it as three "independent" features. Refit, and compare against the original: the AUC, and the fraction of test probabilities above 0.99 or below 0.01. Explain why repeating information makes the model more *confident* but not more *accurate*.

H4. Decision boundaries. On a grid over income and utilisation, plot the $p = 0.5$ contour of Gaussian NB and of logistic regression over the data. One is straight and one is curved. Explain why, using the fact that the two classes have different fitted variances.

Submission

- Call your TA as you finish each task. All three vivas must be done before you leave.
- Keep the notebook; the homework goes in the same file.